

Tatum Beaugh

Data Analyst

Tatumbaugh@gmail.com

Education

Bachelor of Science, Angelo State University
Graduated May 2026.
Major: Computer Science
Minor: Mathematics
Certification: Data science
GPA: 3.1/4.0

Skills

Programming Languages:

- Python, R, C++, Java, SQL

Libraries:

- Matplotlib, pandas, XGBoost, SHAP, tensorflow, Scikit Learn

Other:

- Machine learning, AI, Data science, Calculus, Linear Algebra, Statistics

Awards and Memberships

- Angelo State CS Club, Vice President (January 2023 - December 2024)
- Angelo State University Dean's list (2023 - 2025)

Work Experience

Inventory Data Analyst — Principal Industries (April 2026 - Present)

Ensure accuracy of inventory adjustments, average costing, and unit of measurement within NetSuite ERP system. Maintain and update item attributes for a catalog of over 10,000 SKUs, supporting efficient data governance. Prepare and analyze financial reports related to inventory and transactions for management decision-making. Lead and support small-scale project initiatives to improve operational processes and system performance.

Research

Bridging the Black Box: Explainable Anomaly Detection for Critical Infrastructure - Angelo State University AI-COE

(September 2025 - February 2026)

Utilized knowledge graphs to construct structured, semantic representations of physical and operational data across a complex waterbed management system. Applied explainable AI (XAI) techniques, including SHAP and XGBoost models, to detect, classify, and interpret system anomalies, providing clear diagnostic insights for critical infrastructure monitoring.

Geospatial modeling with FIA data - Purdue University Data Mine

(August 2025 - May 2026)

Partnered with the USDA Forest Service to process satellite imagery and spatial data via Google Earth Engine, predicting ecological indicators in under-reported regions. Applied machine learning models—including random forests and spatial regression algorithms—to analyze regional environmental impacts.

Analyzing Congressional Polarization - Purdue University Data Mine

(August 2024 - April 2025)

Collaborated with C-SPAN to analyze congressional polarization within floor speeches by developing natural language processing and neural network pipelines. Applied advanced text mining, sentiment analysis, and machine learning techniques. Visualized key trends to provide actionable insights into legislative discourse trends.